

Apply filters to SQL queries

Project description

Through SQL we can filter data to find many items such as log in attempts, log in time, user id's, ip addresses and much more. This can help us to find problems or irregularities in a companies log and resolve the issue with more understanding.

Retrieve after hours failed login attempts

Here we search for the failed login attempts that happened before 18:00. First we take the login_time data and make that less than 18:00, then make the success of the login 0 (or false).

```
clear
MariaDB [organization]> select *
  -> from log_in_attempts
  -> where login_time < '18:00' and success = 0;
```

Snippet of the results:

```
+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
| 4 | dkot | 2022-05-08 | 02:00:39 | USA | 192.168.178.71 | 0 |
| 5 | jrafael | 2022-05-11 | 03:05:59 | CANADA | 192.168.86.232 | 0 |
| 6 | arutley | 2022-05-12 | 17:00:59 | MEXICO | 192.168.3.24 | 0 |
| 8 | bisles | 2022-05-08 | 01:30:17 | US | 192.168.119.173 | 0 |
+-----+-----+-----+-----+-----+-----+-----+
88 rows in set (0.001 sec)
```

Retrieve login attempts on specific dates

On this we search the specific date that an issue occurred which was 2022-05-08. Here only data on that date is shown.

```
MariaDB [organization]> select *
  -> from log_in_attempts
  -> where login_date = '2022-05-08';
```

Snippet of the results:

```
+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
```

4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1

28 rows in set (0.033 sec)

Retrieve login attempts outside of Mexico

In the practice example we found an issue that comes from somewhere other than Mexico. In the code we filter out Mexico by putting not and to include all forms of Mexico (Mex and Mexico) we add like and %.

```
MariaDB [organization]> select*
-> from log_in_attempts
-> where country not like 'MEX%';
```

Snippet of results:

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0

144 rows in set (0.033 sec)

Retrieve employees in Marketing

In this we want the data for all the employees that are in

```
MariaDB [organization]> select*
-> from employees
-> where department = 'Marketing' and office like 'East%';
```

Snippet of results:

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267

7 rows in set (0.001 sec)

Retrieve employees in Finance or Sales

In this one we only want employees that are in the Finance or Sales department, this is done with the OR function.

```
MariaDB [organization]> select*  
-> from employees  
-> where department = 'Marketing' and office like 'East%';
```

```
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+  
| 1000 | a320b137c219 | elarson | Marketing | East-170 |  
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |  
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |  
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |  
| 1103 | NULL | randers | Marketing | East-460 |  
| 1156 | a184b775c707 | dellery | Marketing | East-417 |  
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |  
+-----+-----+-----+-----+-----+  
7 rows in set (0.001 sec)
```

```
MariaDB [organization]> select*
```

```
-> from employees
```

```
-> where department = 'Finance' or department = 'Sales';
```

```
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+  
| 1003 | d394e816f943 | sgilmore | Finance | South-153 |  
| 1007 | h174i497j413 | wjaffrey | Finance | North-406 |  
| 1008 | i858j583k571 | abernard | Finance | South-170 |  
| 1009 | NULL | lrodriqu | Sales | South-134 |  
+-----+-----+-----+-----+-----+  
71 rows in set (0.001 sec)
```

Retrieve all employees not in IT

Here we want the logs of all the employees that aren't in IT, we simply do this with the NOT function that excludes those in IT.

```
MariaDB [organization]> select*  
-> from employees  
-> where not department = 'Information Technology';
```

Snippet of results:

```
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+  
| 1000 | a320b137c219 | elarson | Marketing | East-170 |  
| 1001 | b239c825d303 | bmoreno | Marketing | Central-276 |  
| 1002 | c116d593e558 | tshah | Human Resources | North-434 |  
| 1003 | d394e816f943 | sgilmore | Finance | South-153 |  
+-----+-----+-----+-----+-----+  
161 rows in set (0.001 sec)
```

Summary

In summary various functions were used to find specific logs about specific events. I used like to search for words that started with the string to keep data that's similar together. Using AND and OR helped me to narrow specific columns of data. Lastly NOT helped to exclude data or find everything but a certain type.